# Exploratory Data Analysis (EDA)

# Summary Report

## Introduction

This report presents an Exploratory Data Analysis (EDA) of the delinquency prediction dataset. The objective is to understand data structure, identify quality issues, and surface early risk indicators to support downstream predictive modeling.

## Dataset Overview

The dataset contains 500 records, representing Geldium's customers with key attributes relevant to delinquency risk. It includes numerical and categorical features such as income, credit utilization, missed payments, and debt-to-income ratio.

Key dataset attributes:

- Number of records: 500

### Data Types

- **Numerical variables:**
  Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Loan_Balance, Debt_to_Income_Ratio, Account_Tenure
- **Categorical variables:**
  Employment_Status, Credit_Card_Type, Location, Month_1 to Month_6
- **Identifier field:** Customer_ID
- **Binary target variable:** Delinquent_Account

## Missing Data Analysis

### Variables with missing values

- Income: 39 missing values
- Credit_Score: 2 missing values
- Loan_Balance: 29 missing values

### Missing Data Treatment

| Missing Data Issue | Handling Method | One-Line Justification |
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|---|---|---|
| **Income** | Median imputation + missing flag | Median is robust to skewed income distributions, and the flag allows the model to capture risk linked to non-disclosure. |
| **Loan_Balance** | account tenure bucket + missing flag | Median imputation by Missing balances often relate to new or adjusted accounts, and segmentation preserves portfolio structure. |
| **Credit_Score** | Global median imputation | Missing rate is negligible, so a simple median minimizes distortion without adding complexity. |

## 4. Key Findings and Risk Indicators

Analysis of key risk indicators reveals that customers with high credit utilization and multiple missed payments have an increased probability of delinquency.

Key findings:

- Strong correlation between high credit utilization (>50%) and delinquency.
- Customers with 3+ missed payments in the past 6 months have a higher delinquency rate.
- Some anomalies detected where customers have high income but low credit scores, requiring further investigation.

## AI & GenAI Usage

GenAI tools were used to summarize dataset trends, detect missing values, and analyze risk factors. The AI-generated insights were cross-validated against known financial risk benchmarks.

Example AI prompts used:

- 'Summarize key patterns in the dataset and identify missing values.'
- 'Analyze delinquency risk based on payment history and credit utilization.

## Conclusion & Next Steps

This exploratory data analysis (EDA) provided important insights into the quality of Geldium's dataset and key risk factors for delinquency. The analysis revealed missing financial data, clear patterns in credit behavior, and some unusual data points that need further investigation.

### *Key Findings:*

- **Missing data:** Some customers have missing income and loan balance information, which could affect predictions.
- **Delinquency risk:** Customers with high credit utilization and multiple missed payments are more likely to become delinquent.
- **Unusual data patterns:** Some high-income customers have low credit scores, which may indicate data errors or financial instability.

### *Next Steps:*

- Decide the best way to deal with missing income and loan balance values, ensuring that the chosen method does not introduce bias.
- Double-check whether high credit utilization and missed payments remain the strongest indicators of delinquency across different customer groups.
- Look into records where customers have high income but low credit scores to see if there are reporting errors or other explanations.

These findings will help Geldium refine how it assesses risk and prioritizes outreach efforts. The next steps should focus on improving data quality, verifying patterns, and preparing for further analysis.